# Area Under a Curve

Help me understand

Exam code: 9MA0

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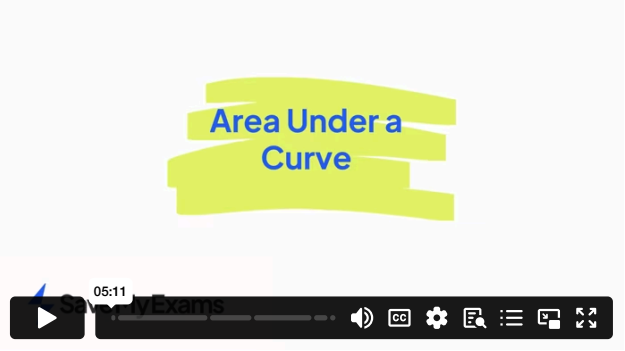


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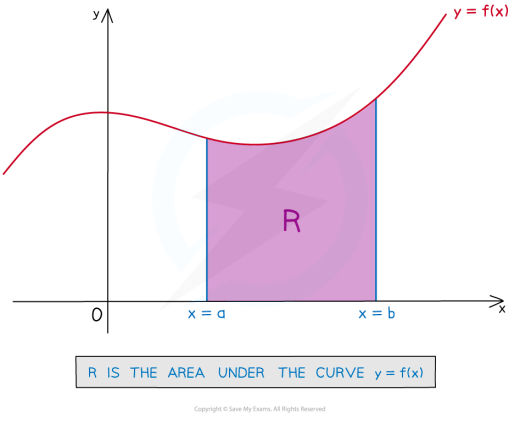
## Area under a curve

### What is the area under a curve?

- The phrase "area under a curve" refers to the area bounded by ...
  - ... the **x**-axis
  - ... the graph of **y = f(x)**
  - ... the vertical line **x = a**
  - ... the vertical line **x = b**

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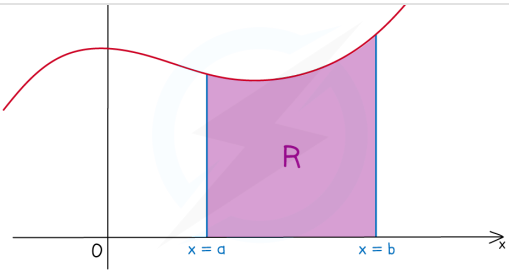
Ask questions and get help with exactly what you're revising.
Got it

### How do I find the area under a curve using definite integration?




$$\text{AREA OF R} = \int_a^b f(x)\,dx$$

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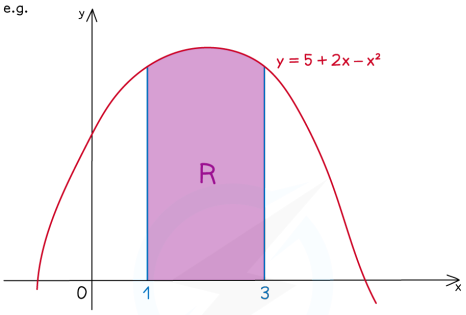
- The value from **definite integration** is equal to the "area under a curve"

e.g.




FIND THE AREA OF THE SHADED REGION.

$$\text{AREA} = \int_1^3 (5 + 2x - x^2)\,dx$$

$$= \left[5x + x^2 - \frac{1}{3}x^3\right]_1^3$$

$$= (15 + 9 - 9) - \left(5 + 1 - \frac{1}{3}\right)$$

$$\text{AREA} = \frac{28}{3} \text{ SQUARE UNITS}$$

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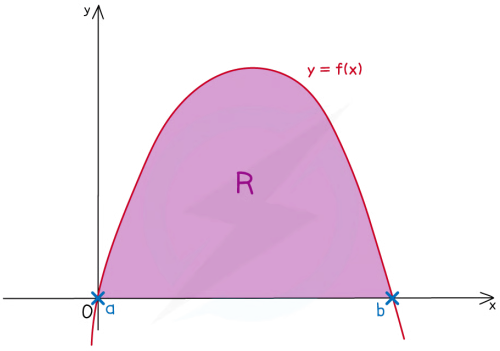
## How do I find the limits of integration?




$$\text{AREA OF R} = \int_a^b f(x)\,dx$$

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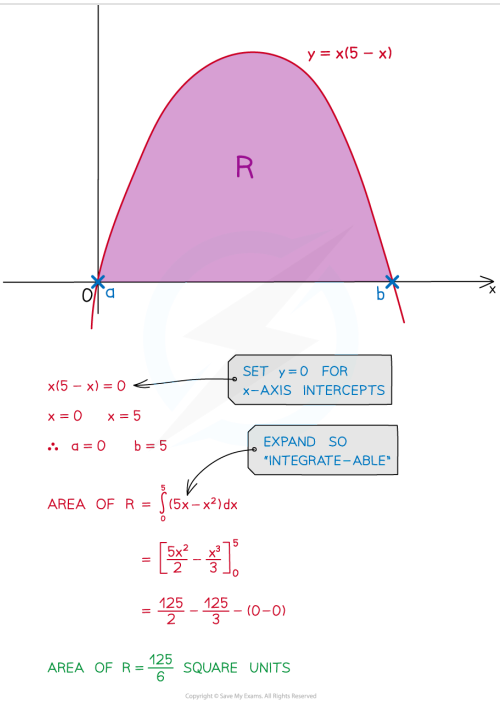
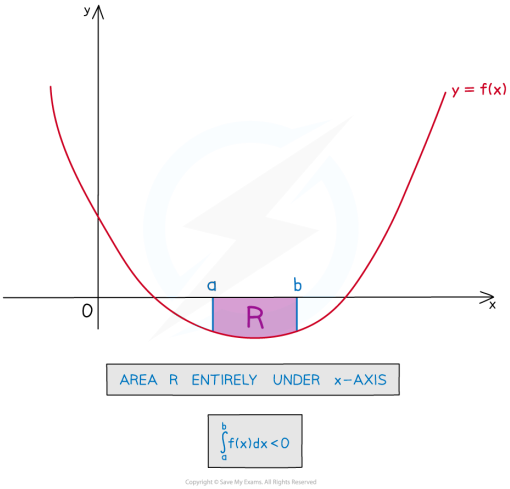
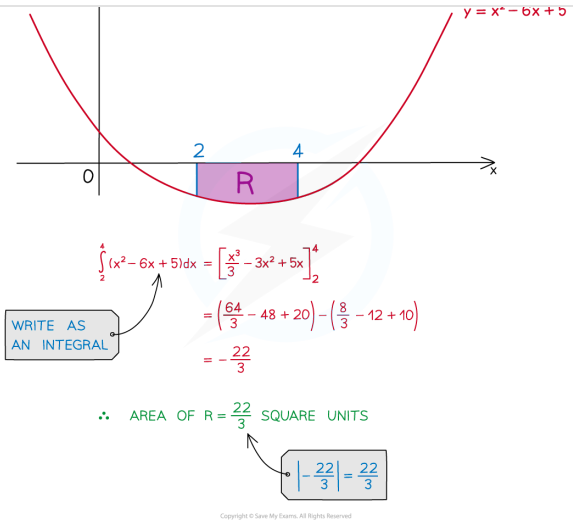
- If limits are not provided they will be the **x**-axis intercepts
  - Set **y = 0** and solve the equation to find the **x**-axis intercepts



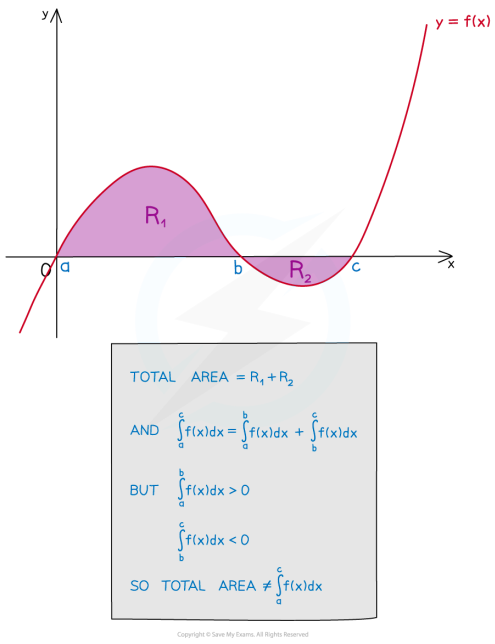

## How do I find areas that are below the x-axis (negative areas)?




- If the area lies underneath the **x**-axis the value of the integral will be negative
- An **area** cannot be negative, so take the **modulus** of the integral




## How do I find total areas without areas cancelling each other out?




- Be careful when one section is above and one section is below the **x**-axis
  - You will need a separate integral for each section, BUT...
  - ...One section's integral will be **negative**
  - ...One section's integral will be **positive**
  - So you'll need to take the **modulus** before adding to find the total area

### Examiner Tips and Tricks

- Add information to any diagram provided in the question, as well as axes intercepts and values of limits.
- Mark and shade the area you're trying to find, and if no diagram is provided, **sketch** one!

### Worked Example

Teacher only

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